Artificial Intelligence

Q: Explain the A* search algorithm in detail with a neat example. Discuss its working, heuristic function, admissibility and advantages and limitations. (10 Marks)

1. Introduction :

- A* search is an informed search technique.
- It finds the optimal (least cost) path from start to goal.
- It uses actual path cost and a heuristic estimate.

2. Evaluation function :

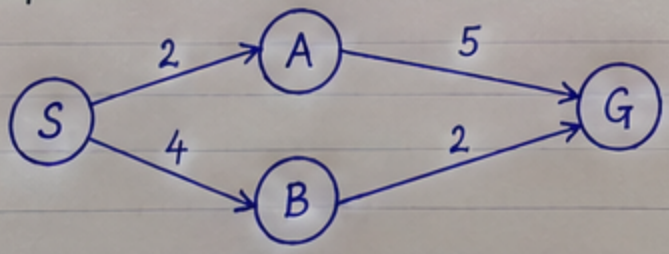
$$f(n) = g(n) + h(n)$$

- $g(n)$: cost from start to current node.
- $h(n)$: estimated cost from current node to goal.
- $f(n)$: total estimated cost.

3. Working of A* :

- Maintain OPEN and CLOSED lists.
- Choose the node with minimum $f(n)$.
- If goal node is reached, return path.
- Otherwise, expand the node and add its successors.
- Update the values and repeat.

4. Example :



Using A*, the shortest path from S to G is selected based on $f(n)$.

5. Heuristic function :

- $h(n)$ estimates the cost from a node to the goal.
- It helps to choose the most promising node.
- Examples : straight line distance, Manhattan distance etc.

6. Admissibility :

- A* heuristic is admissible if it never overestimates the actual cost to reach the goal.
- With an admissible heuristic, A* always finds the optimal path.

7). Advantages :

- Finds optimal solution.
- More efficient than uninformed search.
- Widely used in navigation and game playing.

8. Limitations :

- Requires more memory.
- Performance depends on heuristic function.
- Can be slow for large search spaces.

9. Conclusion :

A* is a useful search algorithm that combines actual cost and heuristic estimate to find the shortest and optimal path.